# From Refining Capacity to Import Dependence: Portugal's Diesel and Gasoline Market, 1990–2024

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**Abstract**

Portugal's refining system gained diesel conversion capability when the Sines hydrocracker entered commercial production in 2013, and lost one of its two refineries when refining ceased at Matosinhos in May 2021. Using Eurostat annual oil balances for Portugal and Spain, JODI monthly physical flows, DGEG national statistics and the European Commission Weekly Oil Bulletin, this paper measures how the product balance and price exposure changed over 1990–2024.

Portuguese demand dieselised as it did across Europe, and for the same fiscal reason [Harding, 2014, Silva et al., 2022]: the continent's refineries were left over-producing gasoline and under-producing diesel [de Boncourt, 2013]. Against that background, Portugal's diesel trade position crosses the same line three times: a net exporter through most of the 1990s, a net importer for the fourteen years from 1999, an exporter again from 2013 in every year but 2019, and an importer in every year since 2021. The 2013 hydrocracker therefore restored a position the country had held before rather than creating a new one, and the 2021 closure carried the system past the dependence it had run in the 2000s. By 2023 domestic manufacturing covers 0.678 of diesel demand and the net-import-to-demand ratio reaches 0.262, the weakest coverage and the heaviest dependence in the window. Gasoline, which entered each event with output far above domestic demand, moves as far on both ratios but crosses neither threshold. On prices, both Portugal–Spain pairs are cointegrated in log levels and both error-correction speeds increase several-fold after May 2021.

No causal effect is claimed. The closure and the March 2022 European supply shock fall ten months apart, and no design here separates their contributions.

**Keywords:** refining capacity; import dependence; petroleum products; energy security; price co-movement; Portugal

## 1 Introduction

Portugal operated two refineries for most of the period studied here. Sines, the larger, received a hydrocracker that entered commercial production in January 2013 and was intended to raise diesel yield relative to fuel oil [Galp Energia, 2013]. Matosinhos ceased refining in May 2021 following a December 2020 announcement, concentrating Portuguese refining at Sines [Galp Energia, 2020]. Ten months later, in March 2022, the withdrawal from Russian oil products disturbed European product supply generally, and vacuum gas oil, a diesel feedstock at Sines, in particular [Galp Energia, 2022].

This paper asks whether the loss of refining capacity increased Portugal's dependence on imported finished petroleum products, and whether Portugal–Spain price co-movement changed after the

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closure. Both questions are answered descriptively and by association rather than causally, for reasons set out in Section 5.

The answer is a sequence rather than a single event, and the sequence is the argument. Portugal was a net diesel exporter through most of the 1990s, in eight of the nine years from 1990. It crossed into net import at the end of that decade and stayed there for fourteen consecutive years. In 2013 the hydrocracker shifts the product slate towards middle distillates and Portugal crosses back into net export, holding that position in every year from 2013 to 2018, with 2019 a single-year interruption and 2020 a brief return. In May 2021 Matosinhos closes and the crossing reverses a third time. Every year since has been a net-import year, and in 2023 the net-import-to-demand ratio reaches 0.262, the highest in the window, while domestic manufacturing covers 0.678 of diesel demand, the lowest.

Read this way the three breaks are not unrelated events but one system crossing and re-crossing a single line. That framing matters for what the 2013 unit did, which was not to move Portugal somewhere new but to restore a trade position the country had held in the early 1990s. And it matters for what followed, because the 2021 closure did not merely undo that restoration. It carried the system past the dependence it had run in the 2000s, to the most import-dependent year in thirty-five.

Three features distinguish the exercise. The annual panel runs from 1990, which is early enough for the 1990s export position to be visible and therefore for the 2013 unit to be read as a restoration rather than a novelty. A panel beginning in 2005, as an earlier version of this analysis did, supports the opposite reading. The monthly design separates the May 2021 closure from the March 2022 shock, which annual data cannot do with ten months between them. And every quantity reported here is checked mechanically against a checksummed evidence bundle before publication, with the checks and their limits described in Section 6.

Section 2 describes the setting and the data. Section 3 defines what resilience is taken to mean here and states the designs. Section 4 reports results. Section 5 interprets them and states what cannot be inferred.

## 2 Setting and data

Four independent bodies publish the series used here. JODI supplies monthly secondary-product imports, exports, refinery output and demand, and carries per-observation assessment codes [Joint Organisations Data Initiative, 2026]. Eurostat's `nrg_cb_oil` supplies the annual oil balance for Portugal and Spain on a common definition, and is the only source reaching back to 1990 [Eurostat, 2026]. DGEG, the Portuguese national statistical authority, publishes annual product trade workbooks from 2019 [Direção-Geral de Energia e Geologia, 2026b] and a domestic sales series from 1970 [Direção-Geral de Energia e Geologia, 2026a]. The European Commission Weekly Oil Bulletin supplies weekly consumer prices with and without taxes from 2005 [European Commission, Directorate-General for Energy, 2026]. Galp corporate disclosures supply dated refinery events. They are used as event metadata only, never as outcome data, and the December 2020 announcement date is deliberately not treated as the physical closure date.

Each arm therefore runs over the study window intersected with what its own source provides: the annual balance over 1990–2024, the monthly flows over 2002–2024 at 276 months per product, and the weekly prices over 2005–2024 at 993 modelled weeks. The bulletin skips weeks: of the 1,044 weeks spanned, 995 are present for Portugal and 994 for Spain, and the gaps are dropped rather than interpolated, so a differenced observation spanning one is a multi-week change entered as a single week.

Diesel is JODI `GASDIES` and Eurostat `O4671`, gas oil and diesel oil. Gasoline is JODI `GASOLINE` and Eurostat `04652`, motor gasoline. Aviation gasoline is excluded from both. Differing biofuel treatment does not drive residual differences between sources, since the Eurostat blended-biofuel wedge is 0.0 kt on exports in every year of the window and at most 4.5% on imports.

Coverage within those windows is complete rather than patchy. Eurostat carries all 280 product-flow cells for both countries from 1990, every JODI year in the window carries twelve observed months, and the price bulletin supplies 995 Portuguese and 994 Spanish weekly observations per product.

One coverage gap is material. DGEG publishes product trade workbooks only from 2019, so for earlier years the trade cross-check runs against Eurostat rather than against the DGEG primary publication. That is weaker corroboration than it appears, because Eurostat's Portuguese oil data is compiled from the DGEG national submission and tracks it to a median difference of 0.00% across the 2019–2024 cells where both exist. Eurostat is best understood as a second channel for the same national statistics, not an independent third opinion. Domestic demand is unaffected, since the DGEG sales series covers the whole window.

Where two genuinely independent sources disagree, both values are reported. Nine of eighty JODI–Eurostat cells and eight of twenty-four JODI–DGEG cells diverge beyond tolerance. No divergent cell enters a headline figure without the alternative value being reported alongside it, and the one case where this changes a conclusion is marked wherever it appears.

# 3 Measurement and design

## 3.1 What resilience is taken to mean

Energy security has no single accepted measure, and the reviews that map the field note that no study operationalises all its dimensions at once [Kruyt et al., 2009, Ang et al., 2015]. Existing quantitative work concentrates on source diversification, political risk and transport risk [Cohen et al., 2011], and the link between import diversification and security is contested rather than settled [Vivoda, 2009]. What follows is narrower than any of these, and deliberately so: it is the part of the concept a national product balance and a bilateral price relation can measure.

Resilience here means the capacity of Portugal's domestic refining and its trade channels to absorb a product-specific supply shock without a sharp increase in import reliance or in exposure to prices formed elsewhere. The definition is deliberately narrow, and the narrowness is why it can be tested, because it asks about the observable response of a physical balance and a price relation rather than about whether supply is secure. Four quantities carry it.

The *domestic supply buffer* is refinery output over domestic demand. This is a capability measure rather than a self-sufficiency measure, since output may be exported while demand is met by imports, which is the normal state for Portuguese gasoline. *Import exposure* is measured both as gross import dependence, imports over demand, and as the net-import-to-demand ratio, which is negative for a net seller and positive for a net buyer. A system can move on one without moving on the other, which is what happens around 2000. *Concentration* is how many places the manufacturing happens in, and Portugal moved from two refineries to one in May 2021. *Price integration* is how closely and how quickly domestic pre-tax prices track a neighbouring market's, which is the price-side counterpart of physical exposure.

For fuel $j$ in year $t$, with imports $M$, exports $X$, demand $D$ and refinery output $Q^{\text{refinery}}$:

$$\text{GrossImportDependence}_{j,t} = \frac{M_{j,t}}{D_{j,t}}, \qquad \text{NetImportToDemandRatio}_{j,t} = \frac{M_{j,t} - X_{j,t}}{D_{j,t}},$$

$$\text{RefineryOutputToDemandRatio}_{j,t} = \frac{Q^{\text{refinery}}_{j,t}}{D_{j,t}}.$$

Several things that belong to a full account of security of supply are absent, and their absence bounds every claim made here. Compulsory stocks and commercial inventory are not observed, and stockholding is the first line of defence against an interruption. Port, storage and pipeline capacity are not observed, nor are supplier diversification, crude slate, contractual terms, or spare capacity at the remaining site. A country importing a large share of its diesel through deep, well-connected markets may be more secure than one manufacturing more of its own from a single vulnerable feedstock route, and nothing here distinguishes those cases.

### 3.2 Designs

Annual breaks are tested by Chow tests on a linear trend at candidate break years, of which 2013 and 2022 were pre-specified and 2000 is exploratory, with 2021 excluded as a transition year so that it is assigned to neither side, and Benjamini–Hochberg adjustment across all twenty-four tests.

The annual interrupted trends are ordinary least squares with a common trend, a post-event level term and a post-event slope term, HAC(1) covariance, 2021 again excluded. This retains all annual observations rather than splitting the series.

Monthly event timing comes from a segmented regression on 276 monthly observations with month fixed effects, a calendar-month trend and HAC(3) errors. Each phase-trend interaction is centred on its phase boundary, so a phase coefficient is the level shift at that boundary measured against the extrapolated pre-closure trend, and the companion trend term is the change in slope within the phase. The two must be quoted together, and a phase coefficient is not an increment on the phase before it.

The retail fuel price literature is largely concerned with asymmetry in how quickly prices rise and fall [Bacon, 1991, Borenstein et al., 1997, Peltzman, 2000], a finding whose robustness has been questioned [Bachmeier and Griffin, 2003] and whose methods have been surveyed with the warning that the field is technique-driven [Meyer and von Cramon-Taubadel, 2004]. No asymmetry is estimated here. The closest published relative runs its diagnostics on euro-area consumer fuel prices from the same bulletin before choosing a specification [Meyler, 2009], and testing a long-run coefficient against a value implied in advance follows Bachmeier and Griffin [2006].

For prices, model family is selected from stationarity and cointegration diagnostics rather than assumed, and the diagnostics are run on log levels because that is the scale the models are estimated on. Testing the EUR per 1000 litres levels instead answers a question about different series, and the two disagree where it matters, since diesel Engle–Granger is 0.057 on the EUR levels and below 0.001 on logs. A levels regression is admitted only on rejection at one per cent rather than five, because regressing one near-integrated series on another is the spurious-regression case while an unnecessary error-correction model still nests the difference specification. Neither product clears that bar, and both pairs are cointegrated in logs, so both are fitted as two-step Engle–Granger error-correction models, estimated with HAC(8) covariance throughout and on pre-tax prices as the primary measure.

## 4 Results

### 4.1 The physical balance

Over 1990–2024, Portuguese diesel demand averaged 4,522 kt against average refinery output of 4,348 kt, average imports of 804 kt and average exports of 643 kt. Gasoline output averaged 2,337 kt against demand of 1,511 kt, a surplus exported at an average of 971 kt a year against imports of only 129.5 kt. Output therefore fell short of demand on average across the window, and the shortfall was covered by buying more than the country sold. For gasoline the arithmetic runs the other way and by a wide margin, since the country manufactured more than it could burn and had to find a buyer for the rest.

That the two products sit in opposite positions relative to the same domestic demand is the single most important fact for reading what follows. A loss of refining capacity acts on a deficit in one product and on a surplus in the other, so the same proportional loss reaches the two balances quite differently. Section 5 takes up why the slate is built that way.

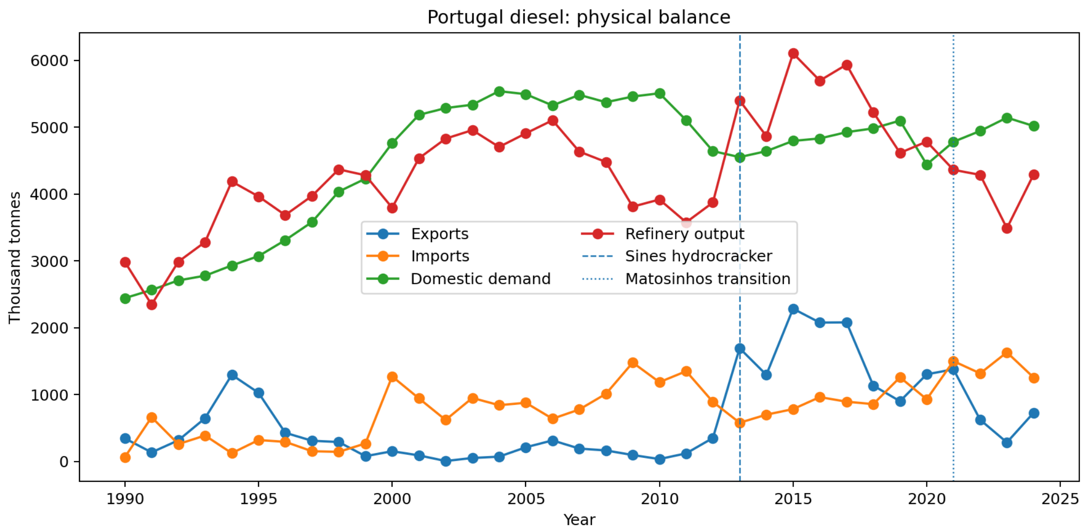


Figure 1: Portuguese diesel imports, exports, refinery output and demand, 1990–2024.

Diesel passes through the four regimes described in Section 1. The line that matters is the refinery-output-to-demand ratio crossing one, because a ratio above one means the country manufactures more diesel than it burns and must place the surplus abroad, while a ratio below one means the shortfall has to be bought.

Taking the later part of the window, from 2005 to 2012 Portugal is a net diesel importer with output covering 0.70 to 0.96 of demand, so the gap was closed by imports in every one of those years. From 2013 the ratio exceeds one and the net-import-to-demand ratio turns negative, which is one event seen from the two sides of the same balance. From 2021 the system returns to net import dependence, and by 2023 output covers only 0.678 of demand, leaving roughly a third of consumption to be sourced abroad.

Gasoline behaves differently. Portugal is a net gasoline exporter in every year of the window except 1993, when imports briefly exceeded exports and the net-import-to-demand ratio reached +0.06. Output covers between 0.94 and 2.66 times domestic demand, the low point being that same year, and from 1994 onward the ratio never falls below one. The ratio rises after 2013 and drifts down after 2021 without approaching one again, so there is no regime change comparable to diesel's.

This is precisely the asymmetry the ratio definition warns about. A system can manufacture a large surplus of one light product while importing another, and reading the output ratio as self-sufficiency would obscure that. It also means the two products cannot be pooled, because an aggregate petroleum-product dependence measure for Portugal would average a structural exporter against a structural importer and describe neither.

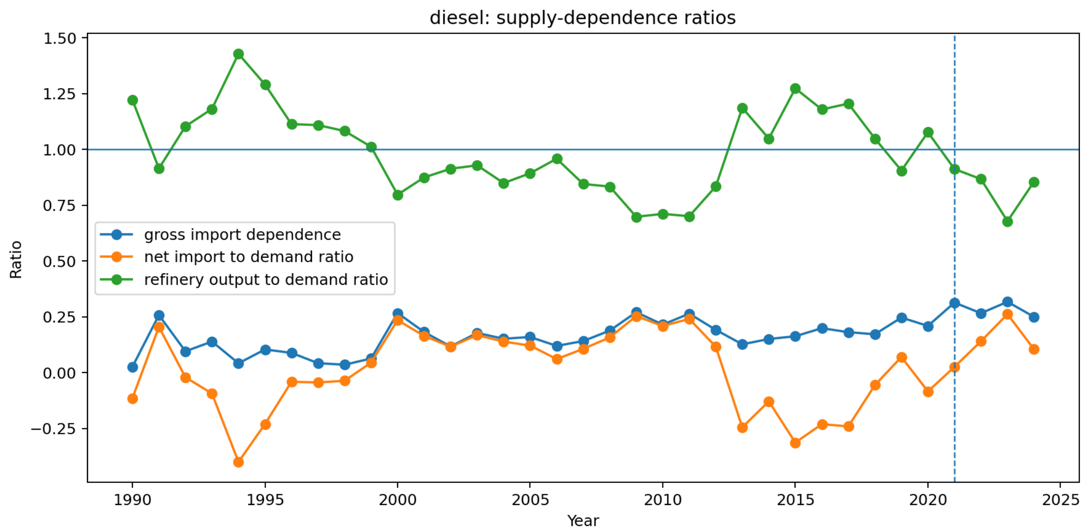


Figure 2: Diesel gross import dependence, net imports to demand, and refinery output to demand. The output ratio crossing one in 2013 and falling back after 2021 is the central physical result of this paper.

Table 1: Balance at the event years. The two independent trade sources disagree on the 2022 diesel export cell. The table carries the national figure of 622.2 kt, which puts net imports over demand at +0.140; the monthly source reports 331.0 kt, which would put it at +0.199. Both values are carried through the supplementary source-sensitivity tables.

| Year | Product | Exports | Imports | Demand | Ref. out. | Net imp./dem. | Regime |
|---|---|---|---|---|---|---|---|
| 2012 | diesel | 350 | 890 | 4,641 | 3,872 | +0.116 | two refineries |
| 2013 | diesel | 1,693 | 578 | 4,551 | 5,399 | −0.245 | two refineries |
| 2020 | diesel | 1,303 | 929 | 4,441 | 4,783 | −0.084 | two refineries |
| 2021 | diesel | 1,377 | 1,500 | 4,780 | 4,361 | +0.026 | transition |
| 2022 | diesel | 622 | 1,316 | 4,942 | 4,285 | +0.140 | Sines only |
| 2023 | diesel | 284 | 1,632 | 5,142 | 3,487 | +0.262 | Sines only |
| 2024 | diesel | 728 | 1,254 | 5,017 | 4,291 | +0.105 | Sines only |
| 2012 | gasoline | 898 | 160 | 1,127 | 1,864 | −0.655 | two refineries |
| 2013 | gasoline | 1,293 | 102 | 1,100 | 2,338 | −1.083 | two refineries |
| 2020 | gasoline | 1,271 | 193 | 883 | 1,956 | −1.221 | two refineries |
| 2021 | gasoline | 1,424 | 202 | 976 | 2,146 | −1.252 | transition |
| 2022 | gasoline | 1,502 | 230 | 1,073 | 2,247 | −1.185 | Sines only |
| 2023 | gasoline | 1,254 | 347 | 1,193 | 2,063 | −0.761 | Sines only |
| 2024 | gasoline | 1,728 | 300 | 1,260 | 2,660 | −1.134 | Sines only |

Table 2: Chow tests at candidate break years, 2021 excluded as a transition year. 2013 and 2022 were pre-specified; 2000 was added after the panel was extended back to 1990 and is exploratory.

| Product | Outcome | Break | $F$ | $p$ | BH $p$ | $n_{\text{pre}}/n_{\text{post}}$ |
|---|---|---|---|---|---|---|
| gasoline | refinery output | 2000 | 22.00 | $< 0.001$ | $< 0.001$ | 10/13 |
| diesel | exports | 2013 | 21.58 | $< 0.001$ | $< 0.001$ | 13/8 |
| diesel | net imports / demand | 2013 | 21.40 | $< 0.001$ | $< 0.001$ | 13/8 |
| diesel | refinery output | 2000 | 11.68 | $< 0.001$ | 0.0030 | 10/13 |
| diesel | refinery output | 2013 | 8.23 | 0.0032 | 0.0152 | 13/8 |
| diesel | imports | 2000 | 5.81 | 0.0108 | 0.0431 | 10/13 |

## 4.2 Structural breaks and event timing

A Chow test asks whether one linear trend describes a series across a candidate year or whether two fitted separately describe it better, so a rejection is a claim about the shape of the series rather than about its level. It says the relationship generating the data changed, not merely that the numbers moved.

Twenty-four tests are run across three candidate break years and six survive Benjamini–Hochberg adjustment, but which series reject matters more than how many. Around 2000 the rejections are confined to refinery output, most strongly for gasoline ($F = 22.0$, adjusted $p < 0.001$), and neither dependence ratio rejects. Output and demand rose together, so the system grew without its trade position moving. A capacity expansion that keeps pace with consumption is invisible to a dependence measure by construction, which is why 2000 is an event in the production series and a non-event in the ratios.

At 2013 the pattern inverts. The rejections reach diesel exports and the net-import-to-demand ratio, both at adjusted $p < 0.001$, and refinery output rejects with them. That combination is what separates the two years. A break in output alone is a statement about what the refineries could make, whereas a break in the ratio is a statement about what the country had to buy, and 2013 is the only candidate year where both move together.

Table 3: Interrupted-trend models, annual series, HAC(1), $n = 34$, 2021 excluded. The 2022 export rows depend on trade cells the two sources dispute, one for each product; the supplementary material re-estimates them on each source. §This row holds the 2013 break constant; the row above fits 2022 alone, and the difference between them is why the annual evidence for a 2022 export break is weak.

| | | Level change | | Slope change | |
|---|---|---|---|---|---|
| Product | Outcome, event | Estimate | $p$ | Estimate | $p$ |
| diesel | exports, 2013 | +2,005.5 | < 0.001 | −117.1 | < 0.001 |
| diesel | exports, 2022 | −781.7 | 0.022 | +15.0 | 0.872 |
| diesel | exports, 2022§ | −570.0 | 0.078 | +150.1 | 0.166 |
| diesel | net imp./dem., 2013 | −0.540 | < 0.001 | +0.029 | < 0.001 |
| diesel | net imp./dem., 2022 | +0.201 | 0.031 | −0.017 | 0.616 |
| gasoline | exports, 2013 | +411.1 | 0.039 | −22.4 | 0.355 |
| gasoline | exports, 2022 | −270.9 | 0.121 | +69.3 | 0.413 |
| gasoline | net imp./dem., 2013 | −0.713 | < 0.001 | +0.038 | 0.157 |
| gasoline | net imp./dem., 2022 | +0.307 | 0.095 | +0.070 | 0.457 |

The two annual designs disagree about 2022, and the disagreement is diagnostic rather than troubling. Chow tests detect nothing on three post-event observations, which is a power statement rather than a null result. The interrupted-trend model, which retains the whole series, detects a level shift in diesel exports of −782 kt ($p = 0.022$). That estimate is considerably weaker on the long series than on a window beginning in 2005, and the gasoline estimates lose significance altogether, so the annual evidence for 2022 is thinner than a short window suggests.

The monthly panel is partitioned into four phases: before May 2021, the Matosinhos transition from May 2021 to February 2022, the energy-stress period from March 2022 to December 2022, and post-stress from January 2023.

Mean monthly diesel imports rise by roughly a half to three quarters between the pre-closure phase and every subsequent phase, refinery output falls by about a fifth in the transition and the stress phase and by nearly a quarter after, and the net-import-to-demand ratio moves from

Table 4: Diesel monthly means by event phase, kt per month unless noted.

| Phase | Months | Imports | Exports | Ref. out. | Net imp./dem. |
|---|---|---|---|---|---|
| Pre-closure | 232 | 75.4 | 65.4 | 411.5 | 0.013 |
| Matosinhos transition | 10 | 133.3 | 63.0 | 321.9 | 0.162 |
| Energy stress 2022 | 10 | 111.1 | 27.7 | 324.6 | 0.193 |
| Post-stress | 24 | 117.5 | 42.3 | 311.7 | 0.181 |

approximately zero to 0.18.

Two features of that pattern matter more than the individual figures. The three terms move in the directions a capacity loss would produce, with imports up, output down and the balance swinging from roughly even to buying about a fifth of consumption abroad. And no phase recovers the pre-closure position, so what the means describe is a level change rather than a disturbance the system worked through. These are raw averages that take no account of seasonality or trend, which the model below does, and a seasonal pattern alone could produce part of a phase difference this size.

Table 5: Segmented monthly event model, diesel, $n = 276$ months (2002–2024), month fixed effects, HAC(3). A phase coefficient is the level shift at that phase boundary against the extrapolated pre-closure trend.

| Outcome | Term | Estimate | Std. error | $p$ |
|---|---|---|---|---|
| Refinery output (kt/month) | Matosinhos transition | −102.89 | 36.53 | 0.005 |
| | phase trend | −4.04 | 6.80 | 0.553 |
| | Energy stress 2022 | −148.25 | 23.41 | $< 0.001$ |
| | phase trend | +5.16 | 3.51 | 0.142 |
| Imports (kt/month) | Matosinhos transition | +40.34 | 23.15 | 0.081 |
| | phase trend | +3.27 | 6.16 | 0.595 |
| | Energy stress 2022 | +25.43 | 10.49 | 0.015 |
| | phase trend | +1.63 | 2.35 | 0.488 |
| Net imports / demand | Matosinhos transition | +0.1923 | 0.0711 | 0.007 |
| | phase trend | +0.0382 | 0.0152 | 0.012 |
| | Energy stress 2022 | +0.3658 | 0.0551 | $< 0.001$ |
| | phase trend | +0.0098 | 0.0072 | 0.171 |

The monthly design separates the May 2021 closure from the March 2022 shock and finds both. Each phase is measured against the same counterfactual, the pre-closure trend extrapolated forward, rather than against the phase before it, so the two coefficients do not add. Diesel refinery output sits about 103 kt per month below that counterfactual during the transition and about 148 kt per month below it during the stress period, with the net-import-to-demand ratio 0.19 and 0.37 above it. The closure phase also carries a significant upward slope of 0.038 per month on the ratio, so dependence continued to widen within that phase rather than stepping once and holding.

### 4.3 The 2022 episode

The chain tested here runs as follows. Russia supplied a large share of the world's vacuum gas oil, which is a feedstock in diesel manufacturing at Sines, and Galp announced the suspension of Russian oil-product purchases in March 2022 [Galp Energia, 2022]. If that disruption bound, diesel manufacturing should fall while gasoline manufacturing does not, because the constraint is on a diesel feedstock and not on running the plant. Domestic demand still has to be met, so exports should give way first and imports rise second. And because Portugal had been a one-refinery system for ten months, none of that adjustment can be taken up domestically.

Each link holds. Against a 2013–2019 baseline, 2022 diesel exports sit at the zeroth percentile and diesel imports of 1,316 kt sit 2.10 standard deviations above the mean, at the hundredth percentile. On the median-and-MAD form, which is less sensitive to the spread of a seven-year baseline, the same figure stands at 2.94. Diesel refinery output sits at $z = -2.04$ and at the zeroth percentile. Gasoline exports are unremarkable at $z = -0.18$, and gasoline refinery output is low but well inside its range at $z = -0.76$. Diesel demand is close to baseline at $z = 0.58$, so the import surge is not demand-driven.

Robust statistics agree. Against the baseline median and median absolute deviation, diesel imports score 2.94 and diesel net imports 3.14, both firmly outside the baseline range. The percentile, which assumes nothing about the shape of the baseline, is the statistic to lean on, and it places 2022 diesel exports at the bottom and diesel imports at the top of seven years of history. The ranking is unchanged across three baseline windows.

Read as a whole rather than statistic by statistic, the four terms of the balance identify where the constraint bound. Refinery output falls to the bottom of its baseline range while gasoline output does not, which locates the problem in diesel manufacturing rather than in running the refinery. Demand sits at its baseline, so nothing was rationed. Exports collapse and imports reach the top of the baseline range. A system that cannot make as much of a product, and whose consumers want the usual amount of it, has two ways to close the gap: stop selling it abroad and start buying it. Portugal did both.

This is what the trade channel absorbing a shock looks like, and it is the only direct observation in this paper of resilience in the sense of Section 3 being tested. The domestic buffer contributed nothing, since output moved the wrong way, which is what a feedstock disruption does. Whether that counts as a system coping or a system exposed depends on the terms on which those imports were obtained, which are not in the data.

2022 diesel exports is also the single largest disagreement between the two independent trade sources, and the reconciliation identifies the monthly source as the outlier for that cell. The qualitative result is unchanged on either value, since every baseline year exceeds both, but the magnitude is not. The standardised deviation is −2.44 on the monthly source and −1.89 on the corroborated national figure. The weaker of the two is the one relied on here.

### 4.4 Portugal against Spain

Table 6: Diesel balance ratios, Portugal against Spain.

| | Refinery output / demand | | | Net imports / demand | | |
|---|---|---|---|---|---|---|
| Year | PT | ES | PT−ES | PT | ES | PT−ES |
| 2018 | 1.049 | 0.899 | +0.150 | −0.056 | −0.046 | −0.010 |
| 2019 | 0.905 | 0.902 | +0.003 | +0.070 | −0.032 | +0.102 |
| 2020 | 1.077 | 0.932 | +0.145 | −0.084 | −0.051 | −0.033 |
| 2021 | 0.912 | 0.847 | +0.065 | +0.026 | +0.009 | +0.016 |
| 2022 | 0.867 | 0.904 | −0.037 | +0.140 | +0.001 | +0.139 |
| 2023 | 0.678 | 0.923 | −0.245 | +0.262 | −0.041 | +0.303 |
| 2024 | 0.855 | 0.898 | −0.043 | +0.105 | −0.040 | +0.145 |

Spain refines the same products, shares a long border across which fuel moves in response to price and tax differences [Leal et al., 2009, Banfi et al., 2005, Teixidó et al., 2024], and shares a product market, and faced the same withdrawal from Russian products on the same timetable. What differs is the event: Portugal closed a refinery in May 2021 and Spain did not. If the post-2021 Portuguese movement were the European shock working through any Iberian refining system, Spain should show something like it.

Over 2018–2024 Spanish diesel output covers between 0.85 and 0.93 of Spanish demand with no trend over those years, and Spanish net imports stay within 0.052 of balance. Portugal covered more of its own diesel demand than Spain did in 2018, 2020 and 2021, and less from 2022, falling to 0.678 in 2023 while Spain sat at 0.923. Portugal reaches 0.262 on the net-import-to-demand ratio in the same year. The comparison is confined to those years deliberately, because across the full panel the Spanish ratio is not flat either, falling to 0.667 in 2007 before recovering, so a claim of stability is defensible only over the window shown.

Spain is a comparison series, not a control. A control would have to be alike in every relevant respect except the treatment, and two national refining sectors never are. Crude slate, refinery maintenance schedules, inventory positions and demand composition all differ, and the VGO disruption did not fall on the two countries equally.

## 4.5 Prices

The order of these tests is not arbitrary, because each one licenses a different model. A series with a unit root wanders without returning to a fixed level, and regressing one such series on another produces a strong-looking relationship whether or not any relationship exists. The first question is therefore whether the price levels are stationary, since a plain regression is only safe if they are. If they are not, the second question is whether the two wander together, which is what cointegration means and what makes an error-correction model the right form.

Augmented Dickey–Fuller tests [Dickey and Fuller, 1979, Said and Dickey, 1984] on the log levels do not reject a unit root for diesel in either country, at $p = 0.123$ for Portugal and $p = 0.119$ for Spain. For gasoline they reject marginally, at $p = 0.019$ and $p = 0.028$, and that verdict is not quite robust: of the six combinations of deterministic term and lag rule tried per series, the Spanish one fails to reject in the case with a trend and lags by BIC, at $p = 0.074$. Across all of them, no series rejects at one per cent. Neither product, then, clears the bar for a levels regression.

Both pairs are cointegrated in logs [Engle and Granger, 1987], diesel at $p < 0.001$ and gasoline at $p = 0.0026$. The two results together settle the specification for both products: the levels wander, they wander together, and an error-correction model is what fits a pair that behaves that way.

### The levels model, and why it is not used

The levels regression was estimated and is retained in the supplementary material, flagged as not valid for inference for both products. It is reported because discarding it silently would hide how much specification choice matters. On that specification the diesel interaction term ES $\times$ post is $-0.082$ ($p < 0.001$), which says that Portuguese prices became *less* responsive to Spanish prices after the transition. The licensed error-correction model gives $+0.662$ ($p < 0.001$) on the same interaction, the opposite sign. The levels model also reports a deterministic trend of $-0.015$ per week ($p = 0.005$). Both features are what a regression between two integrated series estimated without its cointegrating relationship is expected to produce, and the sign reversal is why the stationarity bar is set at one per cent rather than five.

Table 7: Error-correction models, two-step Engle–Granger, HAC(8), $n = 993$ weeks.

| Product | Term | Estimate | Std. error | $p$ |
|---|---|---|---|---|
| diesel | Disequilibrium (lagged) | −0.1415 | 0.0266 | $< 0.001$ |
| | × post | −0.4580 | 0.0845 | $< 0.001$ |
| | post-period level | −0.5995 | 0.0802 | $< 0.001$ |
| | $\Delta$ log ES | 0.7129 | 0.0397 | $< 0.001$ |
| | × post | 0.6624 | 0.0946 | $< 0.001$ |
| | post-period level | 1.3753 | 0.0859 | $< 0.001$ |
| | post | −0.0090 | 0.0021 | $< 0.001$ |
| gasoline | Disequilibrium (lagged) | −0.0911 | 0.0355 | 0.010 |
| | × post | −0.2767 | 0.0859 | 0.001 |
| | post-period level | −0.3678 | 0.0782 | $< 0.001$ |
| | $\Delta$ log ES | 0.7891 | 0.0719 | $< 0.001$ |
| | × post | 0.2846 | 0.2356 | 0.227 |
| | post-period level | 1.0738 | 0.2244 | $< 0.001$ |
| | post | −0.0029 | 0.0017 | 0.092 |

Both long-run relations are close to unit elastic: $\log \mathrm{PT} = 0.193 + 0.970 \log \mathrm{ES}$ for diesel and $\log \mathrm{PT} = 0.123 + 0.978 \log \mathrm{ES}$ for gasoline. The adjustment speed, the share of a gap against that relation closed each week, more than quadruples for both products after the transition. For diesel it moves from $-0.142$ to $-0.600$, shortening the half-life of a deviation from 4.5 to 0.8 weeks. For gasoline it moves from $-0.091$ to $-0.368$, a half-life of 7.3 weeks falling to 1.5. Both changes are precisely estimated.

The two products differ in the other channel. The diesel contemporaneous elasticity rises from 0.713 to 1.375, precisely estimated ($p < 0.001$). The gasoline contemporaneous elasticity does not move on that specification, its interaction being 0.285 with $p = 0.227$. That result does not survive a second break. Tested jointly, a March 2022 regime is rejected for diesel ($p = 0.223$) but accepted for gasoline ($p = 0.020$), and split at both dates the gasoline elasticity runs 0.789 before the closure, 1.679 during the transition and 1.051 afterwards. Gasoline's elasticity is therefore not permanently higher, which is a narrower claim than that it never moved.

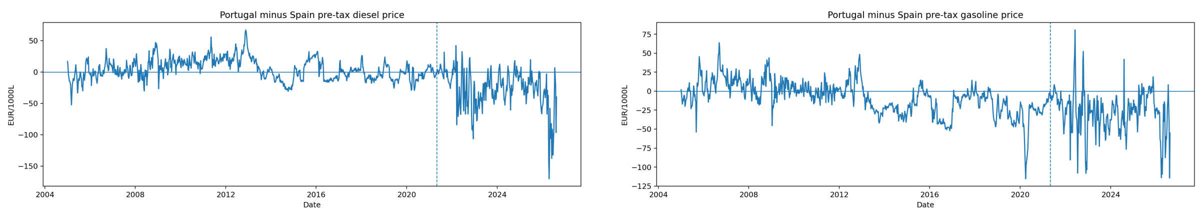


Figure 3: Portugal and Spain weekly pre-tax prices and their spread: diesel (left), gasoline (right).

The mean Portugal-minus-Spain pre-tax diesel spread moves from +4.12 to −26.71 EUR per 1000 litres across the transition, a change of −30.83. The gasoline spread moves by −18.90. These are descriptive. Under test both spreads reject a unit root, on 994 observations each, though not with the same confidence: the gasoline spread rejects decisively (ADF $p = 0.0017$) while the diesel spread clears five per cent and no more ($p = 0.043$). Neither result is a second opinion on the cointegration test, because the raw difference is the cointegrating combination only at a unit coefficient in levels. The estimated slopes are 0.970 and 0.978 and the relation is in logs, so the EUR spread and the modelled residual need not behave alike.

The specification matters for the diesel figure. A difference-only model, which the diagnostics do not licence and which the supplementary material reports, gives a materially lower post-transition elasticity, because omitting the disequilibrium term loads part of the adjustment onto the contemporaneous coefficient. That gap is the reason the model family is selected from diagnostics rather than assumed.

### 4.6 Robustness

Three checks bound how much the results depend on choices made in setting them up.

Table 8: Pre/post difference around 2021 by window, 2021 excluded.

| Product | Outcome | 3 pre / 2 post | 5 pre / 3 post | 8 pre / 3 post |
|---|---|---|---|---|
| diesel | net imports / demand | +0.224 | +0.277 | +0.323 |
| diesel | refinery output / demand | −0.238 | −0.283 | −0.315 |
| gasoline | net imports / demand | +0.333 | +0.393 | +0.281 |
| gasoline | refinery output / demand | −0.388 | −0.424 | −0.321 |

The first varies the pre- and post-event windows around the 2021 transition. The direction of every dependence measure is the same across all three windows and only the magnitudes move, so the headline five-and-three window is not doing the work. On that window diesel gross import dependence rises from 0.202 to 0.278, the net-import-to-demand ratio rises from −0.108 to 0.169, and the refinery-output-to-demand ratio falls from 1.083 to 0.800.

Table 9: 2022 interrupted-trend level changes, computed on each trade source.

| | | Primary source | | Corroborated | |
|---|---|---|---|---|---|
| Product | Outcome | Estimate | $p$ | Estimate | $p$ |
| diesel | exports | −781.7 | 0.022 | −781.7 | 0.022 |
| diesel | net imp./dem. | +0.201 | 0.031 | +0.201 | 0.031 |
| gasoline | exports | −270.9 | 0.121 | −202.8 | 0.149 |
| gasoline | net imp./dem. | +0.307 | 0.095 | +0.253 | 0.111 |

The second re-estimates every 2022 annual result on each of the two trade sources, since the disputed cells fall in that year. One result changes significance, in that the 2022 gasoline export shift does not reach five per cent under either source once the panel is extended, and it is marked wherever it appears and relied on nowhere. The diesel results do not depend on the disputed cell, because the annual panel uses the corroborated source throughout.

The third is multiplicity. Twenty-four Chow tests invite false positives, and a Benjamini–Hochberg correction across all of them leaves six surviving. The 2013 diesel results survive comfortably. The 2022 tests do not survive because they never reached significance to begin with, which is a power statement about three post-event observations.

A fourth check is internal rather than a sensitivity. The Eurostat panel permits a balance identity, $Q^{\text{refinery}} + M - X - D$, that the four reported terms do not close. Across 1990–2024 the Spanish gasoline residual ranges from −20.0 to +15.2 per cent of demand and the Spanish diesel residual from −14.7 to +13.9. The Portuguese ones are smaller but still reach −9.1 and +12.0. Residuals of that size reflect stock changes, refinery fuel and statistical differences, and they are the reason nothing here infers stock behaviour from the balance, since a residual this large would swamp any inventory signal it might carry.

# 5 Discussion

## 5.1 Why the two products diverge

The two products behave differently in almost every result above, and the reason is visible in the window means rather than in anything about how they were treated. Portugal consumes roughly three times as much diesel as gasoline while manufacturing considerably more gasoline than it consumes. Portuguese demand dieselised. Gasoline demand peaks in 2000 and no later year matches it, while diesel demand peaks in 2004, and the gap between the two widened over the window rather than being fixed.

A refinery's products come out of the same barrel in proportions bounded by its configuration, so gasoline output is not an independent decision, because cutting it means cutting throughput, and cutting throughput means making less diesel too. A country running its plants at the rate its diesel demand requires will produce more gasoline than it needs, and the surplus is exported because the alternative is not producing the diesel. The persistent gasoline export surplus is therefore a residue of running the system for diesel, not a strength in gasoline.

That difference in starting position governs the responses. A hydrocracker converts heavy gasoil into middle distillates, so the 2013 unit acted on the deficit product, and diesel output peaks at 6,104 kt in 2015 as the trade position crosses into surplus, while gasoline had no deficit to close and shows no comparable break. The same asymmetry governs the losses. Gasoline entered the 2021 closure with output well above domestic demand, so removing capacity moved the ratio down a wide margin without reaching one, and gasoline coverage stays above 1.7 in every year after the closure. Diesel entered with a ratio near one and no comparable margin, so the same proportional loss pushed it to 0.678 by 2023. A buffer is only protective while it is large relative to the shock, and Portugal held a large one in the product it did not much need and a thin one in the product it did.

This account is consistent with every result above but is not tested by any of them. Yield economics, the crude slate and the commercial logic of export contracts are not observed here, and a competing account resting on contracting rather than chemistry would fit the same series.

## 5.2 What the closure changed beyond capacity

The quantity that changed in May 2021 is capacity. But the closure also changed the shape of the system in ways a capacity figure does not express, and those bear on how the 2022 shock was met. What follows is structural reasoning about a documented event, not an estimate.

With two refineries, an interruption confined to one leaves the other running and the national balance absorbs part of the loss internally. With one, an interruption has no domestic offset at any price and the entire adjustment must run through trade. Turnarounds can be staggered across two sites so that some conversion capacity is always available, and cannot across one. Two plants of different configuration can be run to different slates. One plant's slate is set by its own configuration and the crude available to it.

The consequence that matters most is the concentration of the feedstock exposure. The 2022 disruption was specific to vacuum gas oil, a diesel feedstock at Sines. Before May 2021 a VGO-specific problem at Sines would have left Matosinhos unaffected. After it, Sines was the whole of Portuguese refining, so a shock to one feedstock at one plant was a shock to national diesel manufacturing. The closure did not merely reduce capacity. It concentrated the capacity and its feedstock vulnerability in the same place, ten months before that vulnerability was tested.

## 5.3 What can and cannot be inferred

The causal question is open and this design cannot close it. The transition and the largest European product-supply shock in decades fall ten months apart, too close for annual data and close enough that the monthly design separates them only under an assumption about the shape of the trend between them.

More than that, the two are not independent in a way a better estimator would fix. The closure changed the structure through which the shock propagated, so asking how much of 2022 was the shock and how much was the closure presumes a decomposition the physical system does not have. On that reading the closure and the shock are not competing explanations. The closure determined how a subsequent shock would propagate. The shock determined when that would be tested. Gasoline is the nearest thing to a control the data supplies, having gone through the same closure at the same plant while showing almost none of it in the physical balance, because it entered with a margin diesel did not have. It is not a clean control: its price adjustment speed moved by much the same factor as diesel's.

What follows for policy is correspondingly narrow. The structural exposure measures moved together and in one direction, which a policy discussion would need to start from. But stocks, storage, port capacity, supplier diversity and the terms on which the additional imports were obtained are not observed, and those determine whether higher import exposure is a vulnerability or simply a different and possibly cheaper way of meeting demand. A reader who takes rising import dependence as *prima facie* evidence of reduced security is going beyond what is shown here.

The price finding carries the same limit in a different form. Portuguese pre-tax diesel prices now track Spanish ones more closely and revert to the Iberian relation several times faster, which amounts to a loss of domestic price insulation. That reading sits naturally alongside the physical result, since a country meeting more of its diesel demand from imports is buying at prices formed in the market those imports come from. But it is co-movement between two retail series rather than pass-through, and the specification controls for nothing the two markets have in common, and the same tightening would follow from a period in which common cost shocks came to dominate both. Gasoline is the useful check, and it is the observation least favourable to the framing here, since its adjustment speed moves by a similar factor while its physical position barely changed. Nothing in this paper speaks to refinery-gate or wholesale prices, which are not observed at all.

## 5.4 Limitations

Eight limitations bound the results, in rough order of how much they matter.

1. **2022 confounds the post-closure window.** The transition and the European energy shock fall ten months apart. The monthly model separates them statistically, but COVID-recovery demand, Sines maintenance and the Russian VGO disruption are not separately identified. Two of those run through the concentration the closure produced and are therefore not alternatives to it. Unplanned maintenance is the one that is genuinely independent, and no series here rules it out.

2. **No causal design.** No control group, instrument or counterfactual is used. Every post-2021 result is an association measured around a documented event. The Spain comparison constrains the common-shock explanation but does not license difference-in-differences, and no pre-trend or common-shock diagnostics have been estimated.

3. **Low annual power after 2021.** Three post-event annual observations cannot support a Chow test. The interrupted-trend models carry the annual evidence for 2022, and they

impose a common trend the Chow design does not.

4. **The post-transition period carries the whole price result.** Every interaction in the error-correction models is identified off 190 weeks, a fifth of the sample, and those weeks contain the 2022 price episode. Re-estimating without 2022 and on each half of the period leaves the direction intact and moves the multiple of the pre-transition adjustment speed between 2.96 and 4.40.

5. **Generated regressor in both error-correction models.** The disequilibrium term is estimated in a first stage, so second-stage standard errors are conditional on it [Pagan, 1984]. A single-step or maximum-likelihood estimator would price that uncertainty properly. So does the assumption that one cointegrating vector, estimated over the whole sample, holds on both sides of the transition.

6. **Retail, not wholesale, prices.** The Weekly Oil Bulletin reports consumer prices. The price results describe retail co-movement, not refinery-gate or international wholesale transmission, and are not pass-through in the causal sense.

7. **DGEG trade covers only 2019–2024.** The 2013 break is corroborated against Eurostat, which is compiled from the DGEG submission rather than independent of it.

8. **One cointegrating vector across a break.** The long-run relation is estimated over the whole sample while the adjustment speed is allowed to break. Allowing the relation itself to shift at an unknown date leaves both pairs cointegrated, so the verdict does not depend on the assumption, though the date such a search selects is 2013 rather than 2021.

9. **The Spanish comparison is smaller than its own measurement error.** The Spanish diesel balance residual reaches 15 per cent of demand, against Spanish net imports staying within 0.052 of balance. Spanish flatness is therefore qualitative. The Portuguese movement, about 0.40 on the coverage ratio, is several times its residual and is not affected the same way.

10. **Balance residuals are not closed.** The four Eurostat terms do not sum to zero and stock changes are not modelled.

11. **JODI assessment codes unverified.** All Portuguese observations carry assessment code 1, whose exact semantics have not been confirmed against JODI documentation.

### 5.5 What evidence would settle it

Three post-closure annual observations cannot support a Chow test, and several results here are weak precisely because the post-event window is short. A few more years would let the annual designs speak, and would show whether the 2023 coverage figure was a trough or a level.

Three other kinds of evidence would do more than waiting. A monthly series of Sines throughput and unplanned outages would separate a feedstock disruption from a maintenance event, which annual balances cannot distinguish and which is the most troubling of the remaining alternatives. An import-origin series would show whether the additional diesel came from the markets that set the Spanish price, the mechanism the price result is consistent with but does not demonstrate. And refinery-gate or wholesale quotations would let the price question be asked as pass-through rather than as co-movement.

Failing those, the cleanest natural experiment would be a comparable single-refinery closure elsewhere in Europe at a date not adjacent to a continent-wide supply shock. The identification problem here is not a shortage of Portuguese data. It is that the two events fall ten months apart, and no amount of Portuguese data will move them.

## 6 Conclusion

Over thirty-five years Portugal's diesel system crossed the same line three times and ended further from self-coverage than it began. It was a net exporter through most of the 1990s, a net importer for the fourteen years from 1999, an exporter again from 2013 after the hydrocracker in every year but 2019, and an importer in every year since 2021. By 2023 domestic manufacturing covered 0.678 of diesel demand and the net-import-to-demand ratio stood at 0.262: the weakest coverage and the heaviest dependence anywhere in the window.

Measured against the four senses of resilience defined in Section 3, the profile changed on all four and in the same direction. The domestic buffer thinned, import exposure rose on both measures, manufacturing concentrated from two sites to one, and price integration tightened on both channels. Nothing moved the other way.

Gasoline crossed none of them, which is not the same as saying it did not move. On the annual ratios it moves at least as far as diesel: across the 2021 closure its net-import-to-demand ratio shifts by +0.393 against diesel's +0.277, and its refinery-output-to-demand ratio by −0.424 against diesel's −0.283. Its price adjustment speed quadruples much as diesel's does. What separates them is where they started: gasoline entered every one of these events manufacturing roughly twice its domestic demand and left still manufacturing well above it, so a movement of that size crossed no threshold, while diesel entered with a ratio near one and the same proportional loss carried it across. What determined exposure was not the size of the movement but how much margin the product had before it began.

Three things follow that are not specific to Portugal. The first is that a product-level balance is the right unit of analysis for this question. An aggregate petroleum-product dependence measure would have averaged a structural exporter against a structural importer and shown a system barely moving, when one of its two products crossed from surplus to the heaviest deficit in thirty-five years. The second is that the margin a product holds before a shock, rather than the size of the shock, is what determines whether capacity loss shows up as exposure. The third is that a closure is not only a subtraction of capacity, because it changes the structure through which later shocks propagate, which is why the closure and the 2022 episode resist being weighed against each other.

What the analysis cannot say is how much of the post-2021 position is permanent. Coverage recovers from 0.678 in 2023 to 0.855 in 2024, which is consistent with 2023 having been the trough of an adjustment and equally consistent with a system that now runs closer to its import channel and fluctuates around it. Distinguishing those needs years this panel does not yet have.

## Verification and reproducibility

Every quantity in this paper is generated by a pipeline that writes a checksummed evidence bundle, and the paper is checked against that bundle before it is built. Eleven checks run: each table's printed values must reproduce from the file that table is declared to come from, stated sample sizes must match the fitted models, quoted test statistics must reproduce from columns holding that kind of statistic, no claim may quote a coefficient from a model family the diagnostics did not select, no bundle artifact may cover years the study window excludes, and every quantity carrying a unit must be written where the numeric checks can read it.

Two limits are worth stating, because the checks are easy to over-read. Reproducibility against the whole bundle is close to vacuous for a ratio printed at two or three decimals. The bundle holds hundreds of dense ratio series, and at four decimal places two of six deliberately wrong values still match. Only source-scoped checks discriminate. And no check verifies a claim about a *sequence* of values as opposed to individual ones, which is the class of error that has most often survived review here.

The analysis code, the derived tables behind every figure and table above, the supplementary material referred to throughout, and a document recording the provenance of each input are available in the project repository [Ribeiro, 2026].

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